

Wen-Kai "SansWord" Huang

Senior Software Engineer · Data Infrastructure · Distributed Systems · AI-Assisted Engineering

sansword@gmail.com (669) 220-0281 linkedin.com/in/sansword sansword.github.io/resume San Jose, CA

PROFESSIONAL SUMMARY

Senior Software Engineer with **9+ years** designing and operating large-scale distributed systems and global data infrastructure on **AWS and GCP**, serving **~900M monthly active users** across US, APAC, and Europe. Specialized in high-throughput, fault-tolerant pipelines (>20K msgs/sec), scalable data platforms, and cloud-native migrations. Proven technical leader recognized for cross-functional collaboration and large-scale impact. Since June 2025, on a self-directed build year: **designing, shipping, and publicly measuring LLM-powered systems** and agentic workflows; co-organizer of [AI展 \(aitian\)](#), a weekly AI-application meetup running unbroken since Jul 2026 to **40-50 engineers a session**.

PROFESSIONAL EXPERIENCE

Yahoo!

Sr. Software Dev Engineer · Sunnyvale, CA · Feb 2019 – Jun 2025

- Led end-to-end delivery of Yahoo's data infrastructure evolution to GCP, building a GCP-native ingestion platform on **Cloud Composer, Airflow, and Dataproc** serving US, APAC, and Europe.
- Designed the **per-org, GCP-native pipeline platform** (shared schema module + reusable Terraform template) replacing a centralized ETL shared by orgs including News, Sports, and Finance, so each org owns its own cost and release cycle as Yahoo moved off on-prem data centers; tiered tables cut data scanned **725 GB → 121 KB** and storage/query cost **70%**.
- As tech lead of a 6-engineer team, migrated Sports first, partnering with the Sports team on automated acceptance checks and running old and new pipelines side by side before switching: data freshness **3+ h → 30 min**, new fields in **1 week**; turned a silent-failure incident into job-completion alerting.
- Drove on-call reliability initiative, reducing open incident tickets **from 1,000+ to zero**; practices adopted by the broader on-call team.
- Co-invented a patented salient entity algorithm ([US11803605B2](#)) integrating ranking models into production recommendation systems.

Yahoo! E-Commerce Platform

Sr. Engineer · Taipei, Taiwan · Jun 2016 – Feb 2019

- Drove early AWS adoption across Yahoo engineering, building **blue/green deployment** and auto-scaling patterns via ECS, later adopted by global teams.
- Contributed to migrating **80 TB** to S3 with zero downtime using dual-write/dual-read patterns to ensure safe cutover.
- Scaled **Pigeon** (Apache Pulsar message bus) to **20K msgs/sec**; improved MTBF from 3.5 days to **90 days (~26x)**.
- Core developer of **Cupid**, a multi-tenant discount service with a JSON-based rule engine — driving **10K/day coupon redemption growth**.

PATENT

Determining Salient Entities and Generating Salient Entity Tags Based Upon Articles

[US11803605B2](#)

Co-invented methods to identify salient entities at scale for production recommendation systems at Yahoo.

EDUCATION

M.S., Computer Science

Programming Languages and Software Methodology
National Chengchi University, Taipei, Taiwan

B.S., Mathematics

National Chengchi University, Taipei, Taiwan

COMMUNITY

AI展 (aitian) — Weekly AI Meetup ↗

Co-organizer & moderator, **unbroken since Jul 2026 — 40-50 attendees each (peak 107)**; speakers from AWS, Netskope, Vector Institute, and AI startups. Also built and operate its bilingual portal (CI-validated content, PR contribution flow).

TECHNICAL SKILLS

Languages	Python, Java, SQL, Shell
Cloud – GCP	Dataproc, Cloud Composer, Dataflow, Airflow, BigQuery, GCS
Cloud – AWS	S3, ECS, ELB, ElastiCache
Infrastructure	Terraform, Kubernetes, Docker, Redis
Data Processing	Spark, Hadoop (HDFS), Apache Pulsar, Kafka
AI / LLM	LLM integration (LiteLLM), prompt caching, eval design, guardrails, agentic programming

WRITING & NOTES

sans_blog — Essays & Learning Notes ↗

Field notes on building software with AI — the skeptic-to-shipping story, system-design study guides, and context-engineering deep-dives.

AI ENGINEERING IN PUBLIC

sans_schema

Semantic query gateway — LLM in the request path behind a validated-AST safety boundary; clients query in their own field names + plain English. FastAPI/Postgres, [live demo](#).

guasi (我是)

Live full-stack identity-verification product — Next.js, Postgres/Prisma, NextAuth, preview-deploy CI/CD; built via agentic Claude Code.

sans_bass

Six-stem separation **entirely client-side** — ONNX Runtime + htdemucs_6s on WebGPU at **8.4x realtime**, held to the native pipeline by a parity harness; the audio never leaves the machine. [live demo](#).

sans_cube

Bluetooth Rubik's-cube solve analyzer — BLE, live 3D rendering, CFOP/Roux phase detection.

The Document Tree

Context engineering — a lazy-loaded doc tree keeping an agent's context in sync with the code. Productized as [sans_doc_tree](#).

sans_ai_mock

Multi-mode Claude Code agent for AI mock interviews; measured per-mock API cost.

More work

[sans_learning](#) (7.8x model-cost delta), [cut_sh](#) (ffmpeg scripting), [sans_yt_summary](#) (prompt-injection hardening).